

Encryption in MYSQL

Create table

```
create table employee  
( id int primary key auto_increment,  
name varchar(40), age int ,  
phone varchar(20),  
address varchar(100));
```

Create Stored Procedure normal data insertion

```
DELIMITER $$
```

```
DROP PROCEDURE IF EXISTS sp_insert_employee $$
```

```
CREATE PROCEDURE sp_insert_employee(  
    IN p_name VARCHAR(40),  
    IN p_age INT,  
    IN p_phone VARCHAR(20),  
    IN p_address VARCHAR(100)  
)
```

```
BEGIN
```

```
INSERT INTO employee
```

```
(  
    name,  
    age,  
    phone,  
    address
```

```
)
```

```
VALUES
```

```
(  
    p_name,  
    p_age,
```

```
p_phone,  
p_address  
);
```

```
END $$
```

```
DELIMITER ;
```

Call the Stored Procedure

```
CALL sp_insert_employee('Ramesh',28,'9876543210','Mumbai');
```

Verify now

```
Select * from employee;
```

Create encryption table employee_backup Table

```
CREATE TABLE employee_backup (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARBINARY(255),  
    age INT,  
    phone VARBINARY(255),  
    address VARCHAR(100)  
);
```

Insert Existing Data with Encryption

```
INSERT INTO employee_backup  
(  
    id,name,age,phone,address  
)
```

```
SELECT id,AES_ENCRYPT(name,'Db@d3v2026'),age,AES_ENCRYPT(phone,'Db@d3v2026'),address  
FROM employee;
```

Insert Encrypted Data Using Stored Procedure

```
DELIMITER $$
```

```
DROP PROCEDURE IF EXISTS sp_insert_employee_secure $$
```

```
CREATE PROCEDURE sp_insert_employee_secure(
```

```
    IN p_name VARCHAR(40),
```

```
    IN p_age INT,
```

```
    IN p_phone VARCHAR(20),
```

```
    IN p_address VARCHAR(100)
```

```
)
```

```
BEGIN
```

```
DECLARE encryption_key VARCHAR(100);
```

```
SET encryption_key = 'Db@d3v2026';
```

```
INSERT INTO employee_backup
```

```
(
```

```
    name,
```

```
    age,
```

```
    phone,
```

```
    address
```

```
)
```

```
VALUES
```

```
(
```

```
    AES_ENCRYPT(p_name, encryption_key),
```

```
    p_age,
```

```
    AES_ENCRYPT(p_phone, encryption_key),
```

```
    p_address
```

```
);
```

```
END $$
```

DELIMITER ;

Insert Data

```
CALL sp_insert_employee_secure('Ajit',27,'9876543210','Mumbai');
```

Decrypt Data When Selecting

Stored Procedure to Read Decrypted Data

DELIMITER \$\$

```
DROP PROCEDURE IF EXISTS sp_get_employee_secure $$
```

```
CREATE PROCEDURE sp_get_employee_secure()
```

```
BEGIN
```

```
DECLARE encryption_key VARCHAR(100);
```

```
SET encryption_key = 'Db@d3v2026';
```

```
SELECT
```

```
    id,
```

```
    CAST(AES_DECRYPT(name, encryption_key) AS CHAR(40)) AS name,
```

```
    age,
```

```
    CAST(AES_DECRYPT(phone, encryption_key) AS CHAR(20)) AS phone,
```

```
    address
```

```
FROM employee_backup
```

```
ORDER BY id;
```

```
END $$
```

DELIMITER ;

```
CALL sp_get_employee_secure();
```